

Practice Midterm

Question 1

Check all options that correspond to techniques that can be used to prevent/mitigate model overfitting.

- a) Model regularization
- b) Increase model capacity (e.g., increase the polynomial degree like in the polynomial fitting example seen in class)
- c) Increase the number of samples in the train set
- d) Increase the number of samples in the test set

Answer: a) and c)

Question 2

Check the correct options about cross-validation.

- a) In supervised classification problems, stratified cross-validation is commonly used to maintain the proportions of each class during the cross-validation folds
- b) Cross-validation is commonly used when the dataset available is small
- c) With the advent of deep learning, cross-validation is no longer used
- d) Cross-validation should always be used in any machine learning experiment

Answer: a) and b)

Question 3

Check all the correct questions about the softmax activation.

- a) The softmax and sigmoid activations are equivalent
- b) The softmax activation converts a vector of real numbers to a vector of the same size that can be interpreted as probability scores
- c) The elements of the output vector of the softmax activation can assume any values as long as they sum up to 1
- d) The softmax activation is commonly used at the output layer of multi-label classification problems

Answer: b)

Question 4

Select the metrics that are commonly used for classification problems.

- a) Accuracy
- b) Structural similarity
- c) Sensitivity
- d) Peak signal to noise ratio

Answer: a) and c)

Question 5

Check all options that are correct about regularization in machine learning.

- a) Regularization is never necessary in machine learning
- b) In practice, regularization makes the model parameters assume larger values
- c) If the regularization term is too high, this may cause the model to underfit
- d) Regularization is a technique commonly employed to avoid model overfitting

Answer: c) and d)

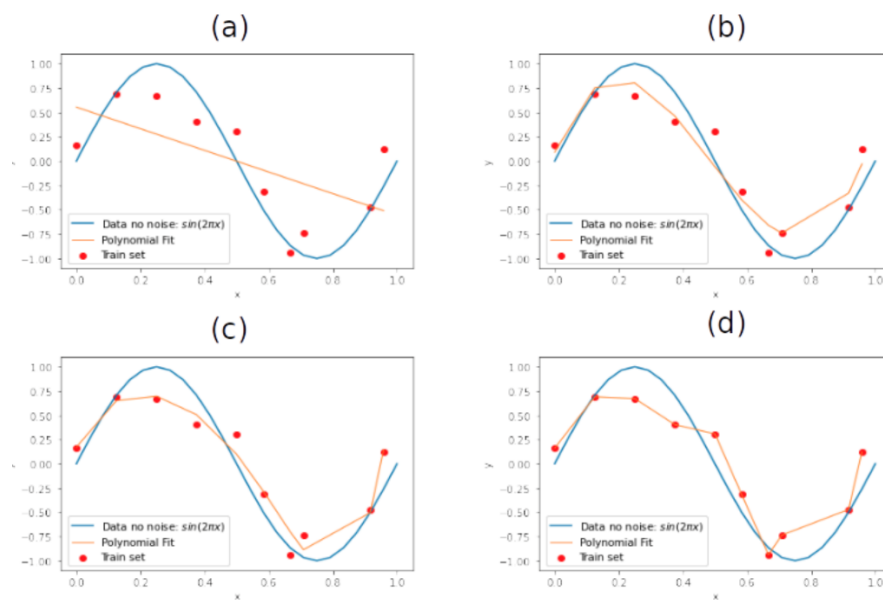
Question 6

A very low error value in the train set is an indication that: (check all options that apply)

- a) The model is definitely overfitting
- b) The model will certainly generalize to new unseen data
- c) The model may be good, but you need to also look at the error in the validation and test sets as well
- d) The model learned something

Answer: c) and d)

Question 7



Regarding the different polynomial fits shown in the figure, check all options that are correct.

- a) The polynomial fit shown in (d) is the best because it fits perfectly all points in the train set
- b) The polynomial fit shown in (a) is a clear case of under-fitting.
- c) The polynomial fits shown in (b) and (c) may represent good fits, but further analysis is necessary.
- d) The polynomial fits shown in (b) and (c) are clear cases of over-fitting

Answer: b) and c)

Question 8

When designing your experiments, you often split your data in a train set, a validation set and a test set. Which of these three sets is used for model selection?

- a) Test set

- b) Validation set
- c) You can use either your validation set or your test set for model selection.
- d) Train set

Answer: b)

Question 9

Check all the correct statements.

- a) Deep learning models commonly alternate linear and non-linear operations
- b) The rectified linear unit (ReLU) is a linear activation
- c) In unsupervised learning, no labels are given to the learning algorithm
- d) Polynomial fitting is a regression problem

Answer: a), c), and d)

Question 10

Check all the correct options about transfer learning.

- a) Transfer learning is only used for classification problems
- b) Models pre-trained on ImageNet are commonly used for transfer learning in image classification problems
- c) Transfer learning relies on the assumption that the representation learned for one problem will be useful for a separate but related problem
- d) In transfer learning to a new task, the top (classifier/regression model) of the originally pre-trained model is replaced by a new top (classifier/regression model) for the new problem you are trying to solve

Answer: b), c), and d)

Question 11

Unlike traditional machine learning, deep learning methods are capable of learning the features directly from the data.

- a) True
- b) False

Answer: a)

Question 12

When training a neural network: (check all the correct options)

- a) Early stopping can be used to save compute by stopping the training if the validation error stops improving for many consecutive epochs (i.e., patience)
- b) It is important to monitor signs of overfitting during training
- c) Your model will always converge to the global minimum of the loss function
- d) Gradient descent is used to maximize the error of your loss function.

Answer: a) and b)

Question 13

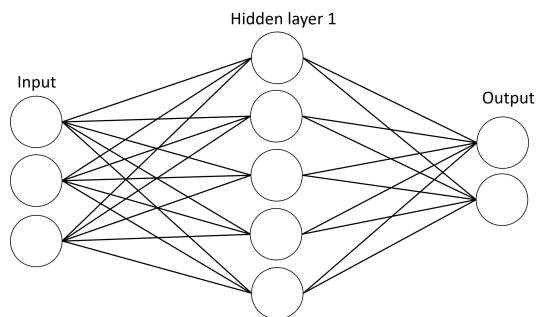
Select the correct options when testing your deep learning model.

- a) The model you select for testing is the one that gave you the highest error in the training set
- b) Gradients don't need to be computed during testing
- c) Dropout layers are often disabled during testing
- d) You should test your model using the training set

Answer: b) and c)

Question 15

In the following fully connected neural network, how many bias parameters are there in total?



Answer: 7 bias parameters

Question 17

Regarding the same network in the previous question, how many parameters are in the first layer?

Answer: $3 \cdot 5 + 5 = 20$

Question 18

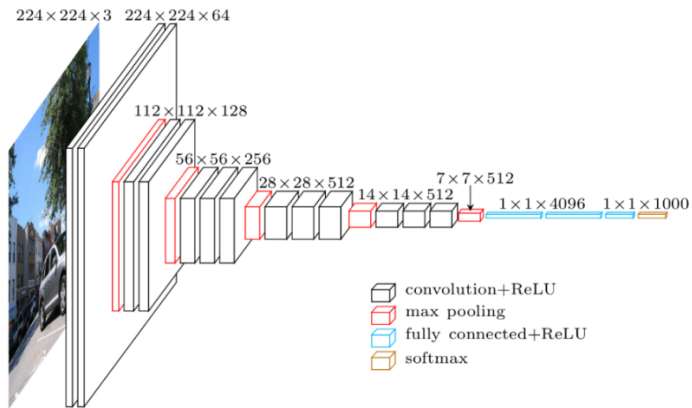
The confusion matrix is an approach for displaying classification results.

- a) True
- b) False

Answer: a)

Question 19

Assuming a 3x3 convolutional kernel, how many parameters are in the first convolutional layer?



Answer: $3 * (3 * 3) * 64 + 64 = 1792$

Question 20

After applying a 3x3 convolutional kernel to a tensor of size 5x5, the resultant tensor will have size:

- a) 3x3
- b) 4x4
- c) 5x5
- d) Not enough information to answer

Answer: d) – we don't know if the convolutions are using padding or not.